

Route Recommendation Method for Frequent Passengers in Subway based on Passenger Preference Ranking

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Abstract: To offer personalized and accurate guidance service to subway passengers, a route recommendation method for frequent passengers in subway based on passenger preference ranking is proposed. Firstly, a data-driven analysis method of travel preference is proposed to identify the travel preference of frequent passengers in different origin-destinations (ODs) and time periods. Secondly, a route recommendation model of dictionary order preference based on just noticeable difference (JND) is proposed to solve the problem of different passenger perception in route attributes. Thirdly, the model of depth deterministic strategy gradient (DDPG) is proposed to consider dynamic parameters of JND. A case study of Guangzhou Metro is illustrated to verify the accuracy of the route recommendation model. The results show that the accuracy of individual passenger recommendation in the proposed method is higher than the traditional logit model and the JND model without parameter optimization by 30% and 20%. The accuracy of group recommendation in the proposed method is higher than the comparative models by 37% and 11%, respectively. Specifically, the accuracy of this method in group recommendation fluctuates less, and its accuracy variance is much smaller than the traditional logit model. Therefore, the method proposed in this study performs better in reflecting personalized passenger travel need.

Keywords: Subway; Route recommendation; JND; Frequent passengers; Passenger preference; DDPG

1 Introduction

With the largescale increase of urban rail transit network and passenger flow congestion, passenger flow control has become important to alleviate passenger flow congestion. There are two ways, direct control which is mandatory evacuation of passengers in the flow restriction zone, and indirect control which sends guidance information to passengers to alleviate passenger crowd indirectly. Most of the literature mainly focuses on the direct control (Xu, Liu, Li, & Jiang, 2016; Xu, Liu, Li, & Hu, 2014). Along with the development of new technologies, the research on the indirect control has become a trend (Lu, Mi, Wang, & Xu, 2021). However, existing route guidance for rail transit passengers does not

consider the characteristics of passenger groups, and cannot offer passengers personalized and precise guidance information. Therefore, it is important to combine individual travel preferences of passengers with the generation of guided routes.

The recommendation of precise and targeted travel routes for passengers needs to identify passenger travel preference at first. In existing research on passenger travel preferences, current focus is the characteristics of travel groups, and in recent years, some research analyses individual passenger travel behaviour. The analysis of individual passenger travel behaviour, especially frequent passengers, is mainly through the establishment of passenger profiles to analyse and characterize individual passenger travel behaviour and preferences (Mao, Guan, & Hao, 2020; Luo, Zuo, Zhang, & Du, 2020; Yu, Li, Xu, Liu, Miao, Wang, & Sun, 2021). Recommend routes is the other work to solve the above recommendation, the main research method is to analyze the factors that influence the passenger's route selection. Usually, a logit model is constructed to recommend routes for passengers (Sun, 2017; Zhang, Yao, & Dai, 2013; Raveau, Guo, Muñoz, & Wilson, 2014; Huang, Zhu, & Yang, 2019; Yao, Chen, Lu, & Yang, 2020; Xu, & Xie, 2022; Zhao, Yan, Yu, & Hentenryck, 2020) based on path scale models, MNL models, NL models, prospect theory and regret theory, where it is difficult to calibrate the model parameters. Currently, the main method for parameter estimation is SP survey which has the shortcomings of insufficient sample size and representativeness. Moreover, most studies only consider the characteristics of passenger groups, and use the same model parameters for all passengers without considering individual passenger preferences.

To fill the gap in this field, this study identifies the route selection preferences of frequent passengers from historical travel records, and studies the heterogeneity of the above preferences for different influencing factors. Based on this, an accurate route recommendation method, that considers passenger perception differences and compares the paths in sequence based on the passenger's preference for different attributes, is proposed.

This study has three main contributions. First, this study proposes a passenger preference identification method based on OD and time period according to historical passenger travel records, and identifies individual passenger travel preferences from the spatial and temporal levels. Secondly, a route recommendation model based on JND's dictionary order preference model overcomes the subjective bias

of the traditional logit model on the weight calibration between attributes, and is more suitable for personalized and dynamic passenger need. Thirdly, this study adopts reinforcement learning to optimize the parameters and automatically calibrate the model parameters that meet the characteristics of passengers from real data instead of questionnaire survey and static fitting in most traditional logit model.

The structure of this study is as follows. Section 2 reviews the related literature on passenger preference and route generation; Section 3 describes the data-driven passenger dictionary order preference recognition method and proposes a route recommendation model based on frequent passenger travel preferences; Section 4 discusses the method and result; Section 5 shows the conclusions and discussion.

2 Literature review

The review of relevant literature includes recent studies on passenger preference and travel behavior (Section 2.1) and an overview of researches on route choice behavior models (Section 2.2).

2.1 Passenger preference and travel behavior

Currently, the analysis of passenger travel behaviour mainly focuses on passenger travel mode and purpose, while the researches on passenger travel preference are relatively rare. Wang, & Lu, (2019) summarized the research methods of passengers' travel as service choice preference. Zhao (2016) and Li et al., (2013) used a questionnaire survey to analyse the behavioural preferences of high-speed rail passengers. Mao et al., (2020) built a group portrait of high-speed railway passengers from seven dimensions based on mobile big data and SP survey analysis. In terms of rail transit. Zou et al., (2018) classified passengers by travel intensity, travel time, spatial distribution, and card type, and analyzed the travel characteristics of different types of passengers. Luo et al., (2020) built a portrait of frequent passengers based on the passenger flow data of Qingdao Metro, and analyzed the travel characteristics of frequent passengers. Yu et al., (2021) fused AFC and point of interest data, and used density clustering algorithm to identify the travel purpose of subway passengers. Long, & Thill, (2015) combined smart card data with household survey data to study the spatiotemporal travel patterns of commuters. At present, most of these studies have analyzed the travel characteristics of groups, but few have studied the travel characteristics of individual passengers. At the same time, for individuals, the number of unusual

passengers traveling by rail transit is relatively small, and it is difficult to extract their travel rules using AFC data. Therefore, frequent passengers that were mentioned in Lu et al., (2012) are mostly taken as the research object.

2.2 Route choice model

In terms of travel route choice considering passenger preferences, current research mainly focuses on influencing factor analysis, route choice model construction, and model parameter calibration. For factors affecting passenger route selection, travel time, transfer time, transfer times, and congestion level are usually considered (Wang, & Jiang, 2020). Shi et al., (2015) analyzed passenger travel time and found that the main factors affecting passenger route selection included walking time, waiting time in departure stations, transfer waiting time and ride time. Ni et al., (2021) added virtual transfer stations to quantify the impact of transfer on the travel route selection of passengers, and constructed a three-layer multi-standard rail transit topology network model without hidden connections. When calculating fixed utility, Liu et al., (2009) added a transfer penalty coefficient as well as transfer times, transfer time and travel time. On the basis of transfer penalty, psychological feelings of passengers are also taken into consideration in the passenger choice model (Si et al., (2007)). Qian et al., (2015) fully considered that transfer time affected passengers' psychological cognition of transfer cost. And the longer transfer time is, the greater passengers' psychological aversion to the route is. Chen et al., (2017) studied train congestion degree as an influencing factor based on passengers' psychological feelings towards travel environment. Batarce et al., (2016) considered the influence of congestion on passengers' routing behavior. Zhu et al., (2018) took into account the influence of passenger retention on routing, when studying passengers' routing behavior. Xu et al., (2018) considered the influence of train load on route selection behavior in urban rail transit. As for this study, the influencing factors include travel time, waiting time, transfer times and route congestion degree.

Many scholars have studied route selection models based on stochastic utility and discrete choice model. Sun, (2017) constructed a MNL model of route choice for urban rail transit passengers. Zhang et al., (2013) considered the influence of network characteristics on travel route selection when establishing a MNL model, and comprehensively analyzed other preferences of passengers when choosing travel routes. On

the basis of considering population factors in social factors, Raveau et al., (2014) used C-logit model to describe passenger route selection. Huang et al., (2019) proposed a route selection model based on prospect theory, whose results were superior to those models based on expected utility theory. In terms of route selection with passenger preference, Zhang et al., (2018) proposed a semi-compensated NL model, where the tolerance threshold of passengers for route selection reflected passenger heterogeneity. Yao et al., (2020) proposed an NL model considering rational decision-making and inherent choice preference, which was superior to the traditional NL model; Xu, & Xie, (2022) established a mixed utility-regret model integrating the guidance information and taking into account the tendency difference of passengers to follow the mixed decision rules and the perception diversity for different attributes when making decisions.

Although the above models partly consider passenger preferences, they do not consider the degree of passenger preferences for different influencing factors of route choice. Jiang et al., (2021) proposed a novel stochastic transit assignment model that considered personalization and bounded rationality. By Lexicographic order sorting travel time, waiting time and congestion cost, passengers' travel preferences were considered to achieve personalized traffic assignment. However, Lexicography ignores passengers' perception difference, that is, if the difference of a certain attribute between the two paths is within a certain range, passengers cannot determine that there is a difference between the two paths. To solve this problem, Zhong et al., (2022) proposed a novel α -conditional lexicographic optimization method to capture the decision-maker's perceived acceptability of the trade-off between different objectives with respect to the hierarchical objective ordering. Ceder et al., (2020) developed a novel, lexicographical shortest path method, considering a just noticeable difference (JND). The method adopted lexicographical ordering to capture passenger preferences for different attributes of routes following Ernst Weber's law of human perception threshold (Ceder, & Jiang, 2020). Therefore, this study proposes a lexicography preference model based on JND to consider the influence of passenger preferences on route selection. In terms of model parameter calibration, most of the existing studies use survey statistics, and there is a problem of small sample size and insufficient representation. In addition, most studies only consider the

characteristics of passenger groups, and adopt the same model parameters for all passengers without considering individual preferences of passengers.

3 Methodology

3.1 Data-driven lexicographic recognition of passenger preference

In this section, based on AFC, historical load rate, platform waiting time and other multi-source data, passenger lexicographic preference degree label is constructed to analyze passenger's travel preferences. Furthermore, a passenger preference recognition method based on OD and time period is proposed to identify passenger preferences from both spatial and temporal aspects.

Lexicographical passenger preference refers to the degree of preference to route attributes when passengers choose travel route. The route attributes include travel time, wait time, transfer time and congestion degree (Sun, Shuai, & Huang, 2019). In the model, one passenger is defined by c and Q^c stands for the lexicographical preference set. Therefore, passenger preference identification for each route attribute is determined as shown in Eq. (1).

$$Q^c = \{q_1, q_2, q_3, q_4\}, q_i \in \{\text{travel time, wait time, transfer time, congestion degree}\} \quad (1)$$

where the higher rank of q_i has the higher the degree of passenger preference. The elements of Q^c correspond to the relation shown in Eq. (2).

$$q_i > q_{i-1} \quad \text{or} \quad q_i = q_{i-1} \quad (2)$$

where $q_i > q_{i-1}$ means the preference of q_i is prior to q_{i-1} ; $q_i = q_{i-1}$ means the preference relation of q_i and q_{i-1} is equal. Considering the availability of data and the characteristics of existing data, this study takes passengers who travel more than 3 times a week (frequent travelers (Lu, Han, & Cai, (2012))) as the research object.

Frequent travelers traveling in the same OD and time period often show the same route selection preference. Therefore, this study divides frequent travelers from both spatial and temporal aspects, makes cluster analysis on frequent travelers traveling in the same OD and time period, and automatically determines passenger categories and identifies passenger preferences.

The characteristics of the route selected by passengers reflect the travel preferences of passengers. Therefore, clustering indicators include average travel time, average waiting time, average transfer times and average congestion degree of the route. However, the above four indicators can only reflect the route characteristics but can't reflect the factors of passenger route selection. Passengers prefer to choose the route that satisfies own preferences and conforms to perceived difference (Xu, & Xie, 2022). Therefore, to reproduce the factors considered by passengers when choosing routes, different attribute labels for each route are established by analyzing the travel route selected by passengers. Different attributes correspond to labels such as "least travel time", "least wait time", "least transfers" and "least crowdedness". The label establishment process is as follows.

Let x^c be the record of passenger c choosing route k from start station o to terminal station d at time t_{in} in tp period, and x^c constitutes the whole travel record set X^c . Let $L^{o,d}$ be the feasible route set of travel record (o, d) . For route $l_k^{o,d}$ in $L^{o,d}$, route attribute $y_k^{o,d,t_{in}}$ at time t_{in} can be calculated as follows:

$$x^c = (c, t_{in}, t_{out}, o, d, k, tp) \quad (3)$$

$$y_k^{o,d,t_{in}} = (t_k^{o,d,t_{in}}, wt_k^{o,d,t_{in}}, tr_k^{o,d}, z_k^{o,d,t_{in}}) \quad (4)$$

where $t_k^{o,d,t_{in}}$ represents the travel time of route k from start station o to terminal station d ; $wt_k^{o,d,t_{in}}$ represents the platform waiting time of route k from start station o to terminal station d at time t_{in} ; $tr_k^{o,d}$

represents the transfer time of route k from start station o to terminal station d ; $z_k^{o,d,t_{in}}$ represents the congestion degree of route k from start station o to terminal station d at time t_{in} .

The congestion degree of travel route is measured by the proportion of crowded intervals in all routes. Considering the influences of congestion degree, this study uses the weighted proportion of crowded interval as the index to measure the congestion degree of travel route, as shown as follows.

$$z_k^{o,d,t_{in}} = \frac{\sum_{t_{ij} \in W_k^{o,d}} t_{ij} \times \lambda}{\sum_{t_{ij} \in W_k^{o,d}} t_{ij}} \quad (5)$$

$$\lambda = \begin{cases} 0 & c_{ij}^{t_{in}} \in [0, 80\%) \\ 1 & c_{ij}^{t_{in}} \in [80\%, 100\%) \\ 1.1 & c_{ij}^{t_{in}} \in [100\%, 120\%) \\ 1.2 & c_{ij}^{t_{in}} \in [120\%, +\infty) \end{cases} \quad (6)$$

where $W_k^{o,d}$ represents the interval set of route k from start station o to terminal station d ; λ represents the weight corresponding to the congestion degree of interval [35]; $c_{ij}^{t_{in}}$ represents the full load rate of the interval at time t_{in} .

Let $Y_{o,d}^{t_{in}}$ be the set of attributes for each route of $L^{o,d}$ at time t_{in} . After comparing the attributes of each route, the attribute label $tag_{q_i}^{o,d,k,t_{in}}(q_i \in Q_c)$ of route k at time t_{in} is established. The value of the label is 0 or 1. The value 0 indicates that the route does not have the label feature, while the value 1 indicates that the route has the label feature. Further the degree of passenger preference for each label is calculated. $p_{q_i}^c$ represents the preference of passenger c for label q_i , and its calculation method [36] is shown as follows.

$$p_{q_i}^c = \frac{f_G^{c,q_i}}{f_G^{c,q_i}(o,d)} \times 100 \quad (7)$$

$$f_G^{c,q_i} = \frac{\sum_{(o,d,k,t_{in}) \in X^c} tag_i^{o,d,k,t_{in}}}{|X^c|} \quad (8)$$

$$f_G^{c,q_i}(o,d) = \frac{\sum_{(o,d) \in L^{o,d}} tag_{q_i}^{o,d,k,t_{in}}}{\sum_{(o,d) \in X^c} |L^{o,d}|} \quad (9)$$

where f_G^{c,q_i} represents the proportion of the travel routes that contain label q_i chosen by passenger c ; $f_G^{c,q_i}(o,d)$ represents the proportion of the feasible route set of travel OD that contains label q_i .

In conclusion, the clustering index selects average travel time, average waiting time, average transfer time, average congestion degree, least travel time label, least waiting time label, least transfer label, and least crowdedness label of travel routes. The clustering algorithm selects spectral clustering and uses contour coefficient and CH fraction to determine the number of clustering. Finally, passenger preference Q^c of passenger c is demonstrated by the clustering center of its category.

3.2 Route recommendation model based on frequent traveler's preference

In this section, passenger preference characteristics are integrated into the route recommendation model, and the JND-based lexicographical preference model is proposed to match passengers and routes. Furthermore, passenger sensitivity to the change of route attributes is considered, and a model parameter optimization method based on reinforcement learning is proposed to make the matching more accurate and personalized.

In route comparison, the JND-based lexicographic preference model firstly compares the attribute with the highest degree of route preference. If the attribute difference can be perceived by passengers, passengers are considered more willing to choose the route with this attribute; otherwise, the two routes

are considered no difference in this attribute, so the second-highest attribute needs to be compared (Ceder, & Jiang, 2020).

Let $\hat{Y}^{o,d,\omega}(Q)$ be route attribute set of the OD at time ω . And passenger perception for attribute changes is assumed to obeys Weber's law, where there is JND threshold. Let $\beta_{q_i}^{JND}(q_i \in Q^c)$ be the perceived change ratio of passengers for each attribute change. The objective of the recommendation model is to determine the optimal route l_{best} , which satisfies passenger preference. The main idea of this model is to use $\hat{Y}^{o,d,\omega}(Q)$ for critical condition, and sort $L^{o,d}$ which is the OD route set to obtain an ordered route set satisfying passenger preference. Furthermore, the more advanced the route in $L_{fav}^{o,d}$ is, the more satisfied the preference of passengers is. Therefore, the first ranked route $l_{fav}^{o,d,1}$ in $L_{fav}^{o,d}$ is defined as l_{best} .

Specifically, the following rules are obeyed during the ranking process as follows.

If the following conditions are true, route $l_k^{o,d}$ is superior to route $l_p^{o,d}$ for passengers at time ω .

$$l_k^{o,d} > l_p^{o,d} \Leftrightarrow \begin{cases} \exists q_i \in Q^c, y_{p,q_i}^{o,d,\omega} > y_{k,q_i}^{o,d,\omega} \text{ and } y_{p,q_i}^{o,d,\omega} - y_{k,q_i}^{o,d,\omega} > \beta_{q_i}^{JND} \cdot y_{q_i}^{o,d,\omega} & (a) \\ \forall m \leq i-1, \begin{cases} y_{j,q_m}^{o,d,\omega} - y_{q_m}^{o,d,\omega} \leq \beta_{q_m}^{JND} \cdot y_{q_m}^{o,d,\omega}, \forall j \in \{k, p\} & \text{or} \\ y_{p,q_m}^{o,d,\omega} = y_{k,q_m}^{o,d,\omega} \text{ and } y_{j,q_m}^{o,d,\omega} - y_{q_m}^{o,d,\omega} > \beta_{q_m}^{JND} \cdot y_{q_m}^{o,d,\omega}, \forall j \in \{k, p\} & (b) \end{cases} \end{cases} \quad (10)$$

where $y_{q_i}^{o,d,\omega}$ represents the minimum value of attribute q_i in the OD route set and $y_{i,q_i}^{o,d,\omega}$ represents the value of attribute q_i in route i of OD (o,d) at time ω . There are two possible conditions. The first one is that the value of route $l_p^{o,d}$ is greater than that of route $l_k^{o,d}$, and their difference is greater than the JND threshold for attribute q_i . The other is that either the difference between the two routes is within the JND threshold for the first $i-1$ attributes or the two routes have the same attribute value and the difference between the value of one route of attribute q_i and the minimum value is greater than the JND threshold.

If the difference between the two routes is within the JND threshold for the all attributes or the two routes have the same attribute value and the difference between the value of one route of attribute q_i and the minimum value is greater than the JND threshold, there is no difference between route $l_k^{o,d}$ and route $l_p^{o,d}$ when passengers choose one route at time ω , as shown Eq. (11):

$$l_k^{o,d} = l_p^{o,d} \Leftrightarrow \forall q_i \in Q^c, \begin{cases} y_{j,q_i}^{o,d,\omega} - y_{q_i}^{o,d,\omega} \leq \beta_{q_i}^{JND} \cdot y_{q_i}^{o,d,\omega}, \forall j \in \{k, p\} & \text{or} \\ y_{k,q_i}^{o,d,\omega} = y_{p,q_i}^{o,d,\omega} & \text{and } y_{j,q_i}^{o,d,\omega} - y_{q_i}^{o,d,\omega} > \beta_{q_i}^{JND} \cdot y_{q_i}^{o,d,\omega}, \forall j \in \{k, p\} \end{cases} \quad (a) \quad (b) \quad (11)$$

The calculation process of the JND-based lexicographic preference route recommendation model is as follows:

Step 1: An ordered preference attribute set Q^c , the perception change ratio $\beta_{q_i}^{JND}$, and the route set matrix of OD $M^{o,d}$ are input to the model.

$$M^{o,d} = \begin{bmatrix} y_{1,q_1}^{o,d,\omega} & y_{2,q_1}^{o,d,\omega} & \cdots & y_{n,q_1}^{o,d,\omega} \\ y_{1,q_2}^{o,d,\omega} & y_{2,q_2}^{o,d,\omega} & \cdots & y_{n,q_2}^{o,d,\omega} \\ \cdots & \cdots & \cdots & \cdots \\ y_{1,q_n}^{o,d,\omega} & y_{2,q_n}^{o,d,\omega} & \cdots & y_{n,q_n}^{o,d,\omega} \end{bmatrix} \quad (12)$$

Step 2: Initialize the recommended route set $\hat{L}_{fav}^{o,d}$ and temporary route set $\bar{L}^{o,d}$, where $\hat{L}_{fav}^{o,d}$ is the empty set and $\bar{L}^{o,d}$ is the orderly route set, from smallest to largest ordering based on the attribute value of q_i .

Step 3: Compare the attribute values in the feasible route set in the order of preference, namely traverse the matrix from top to bottom in the order of rows.

Step 4: Compare $Y_{q_i}^{o,d,\omega} = \{y_{1,q_i}^{o,d,\omega}, y_{2,q_i}^{o,d,\omega}, \dots, y_{n,q_i}^{o,d,\omega}\}$. Sort $Y_{q_i}^{o,d,\omega}$ from the smallest to the largest and calculate the upper limit of perceived difference $\varphi_{q_i}^{o,d,\omega}$.

$$\varphi_{q_i}^{o,d,\omega} = \min_{l_n^{o,d} \in L^{o,d}} \{y_{n,q_i}^{o,d,\omega}\} + \beta_{q_i}^{JND} \cdot \min_{l_n^{o,d} \in L^{o,d}} \{y_{n,q_i}^{o,d,\omega}\} \quad (13)$$

If $\bar{L}^{o,d}$ is not empty, take out routes in $\bar{L}^{o,d}$ that is greater than $\varphi_{q_i}^{o,d,\omega}$, and sort them from the largest to the smallest in the $\hat{L}_{fav}^{o,d}$; otherwise, skip to Step 6.

Step 5: After traversing the matrix, if $\bar{L}^{o,d}$ is empty, skip to Step 6. Otherwise, take out routes in $\bar{L}^{o,d}$, and then sort from the largest to the smallest in the $\hat{L}_{fav}^{o,d}$.

Step 6: Rearrange $\hat{L}_{fav}^{o,d}$ in reverse order to obtain $L_{fav}^{o,d}$. The first element of $L_{fav}^{o,d}$ is used as the model output of calculation results.

According to Weber's law, people have different JND thresholds for the same thing, and person also has his own JND thresholds for different things. Therefore, perception change ratio set β should be different for each passenger in the JND-based lexicographic preference model. In this study, reinforcement learning algorithm is used to adjust model parameters, and adapt to the personalization of passenger travel route choice, to improve the accuracy of recommendation route. The main elements of reinforcement learning are as follows:

(1) State

To reflect the accuracy of route recommendation for passengers in parameter optimization, the state of the model is the recommended accuracy δ , namely the ratio of the choice times of recommended route accounting for total travel times. The state space is 1.

(2) Action

The action of the model is $\beta_{q_i}^{JND}$, which is the perception change ratio of the JND-based lexicographical preference model. There are four parameters, so the number of action spaces is 4. The value range of each parameter is [0,1]. The algorithm automatically selects parameters within the value range for learning, avoiding the problem that the learning effect deteriorates when the value interval is

manually determined. Meanwhile, the algorithm automatically explores the parameters, which increases the possibility of parameter selection and is more conducive to the determination of optimal parameters.

(3) Reward function

The reward function of the model is the accuracy difference before and after taking the action. When an action is taken, if the accuracy increases, the reward function is positive, and the agent receiving positive feedback will learn in this direction. If the accuracy decreases, the reward function is negative, and the agent receiving negative feedback will avoid learning in this direction.

DDPG algorithm is used to calibrate parameters. The core of DDPG algorithm is to use deep neural network simulation function and train it with deep learning method (Lillicrap, Hunt, Pritzel, Heess, Erez, Tassa, Silver, & Wierstra, 2015). DDPG mainly consists of two modules, Actor module and Critic module (Wu, & Jin, 2023).

The Actor module is used to update policy functions and select actions to determine an optimal strategy that maximizes cumulative rewards. The parameter of the Actor module is defined as θ^a , and the objective function is shown in Eq. (14).

$$J_{\beta}(\mu) = \int_S \rho^{\beta}(s) Q^{\mu}(s, a) ds = E_{s \sim \rho^{\beta}}[Q^{\mu}(s, a)] \quad (14)$$

$$Q^{\mu}(s_t, a_t) = E[r(s_t, a_t) + \gamma Q^{\mu}(s_{t+1}, \mu(s_{t+1}))] \quad (15)$$

where $Q^{\mu}(s, a)$ is the accumulated reward expectation; ρ^{β} is the distribution function of state s ; a_t is the action at time t ; s_t is the state at time t ; $r(s_t, a_t)$ is the reward value obtained after executing a_t in state s_t ; γ is the decay coefficient of the reward value for the next state, $\gamma \in [0, 1]$.

The Critic module is used to evaluate the current strategy, and its output is the $Q^{\mu}(s, \mu(s))$ in the Actor module. The parameter of the Critic module is defined as θ^Q , and the objective function is shown in Eq. (17).

$$y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1} | \theta^{\mu'}) | \theta^Q) \quad (16)$$

$$L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i | \theta^Q))^2 \quad (17)$$

where y_i is the target Q value; L is the loss function of Critic module.

To improve the stability and efficiency of the algorithm, DDPG creates two neural networks, the main network and the target network, for Actor module and Critic module respectively for training calculation and parameter update. The solution process of the DDPG is as follows:

(1) Initialization

Randomly initialize the main network parameters θ^μ and θ^Q . Initialize the target network parameters $\theta^{\mu'}$ and $\theta^{Q'}$ so that $\theta^{\mu'} = \theta^\mu$ and $\theta^{Q'} = \theta^Q$. Initialize the experience pool R and preset iterations.

(2) Train

Step 1: Initialize a random noise ε_t and obtain state s_t .

Step 2: Select action a_t based on the output of the actor network and noise ε_t , as shown in Eq. (18).

$$a_t = \mu(s_t | \theta^\mu) + \varepsilon_t \quad (18)$$

Step 3: Execute action a_t , receive reward r_t and new state s_{t+1} . Step 4: The actor network stores (s_t, a_t, r_t, s_{t+1}) as a set of data into the experience pool R as the dataset for training the network.

Step 5: Randomly sample N sets of data from R as training data for the actor main network and critic main network.

Step 6: Calculate the gradient of the critic main network and update it with the Eq. (17).

Step 7: Calculate the gradient of the actor main network and update it with the Eq. (19).

$$\begin{aligned} \nabla_{\theta^\mu} J_\beta(\mu_{\theta^\mu}) &= \int_S \rho^\beta(s) \nabla_{\theta^\mu} \mu_{\theta^\mu} Q^\mu(s, a) |_{a=\mu_{\theta^\mu}} ds = E_{S \sim \rho^\beta} [\nabla_{\theta^\mu} \mu_{\theta^\mu}(s) Q^\mu(s, a) |_{a=\mu_{\theta^\mu}}] \\ &= \frac{1}{N} \sum_i \nabla_a Q(s, a) |_{s=s_i, a=\mu(s_i)} \cdot \nabla_{\theta^\mu} \mu(s | \theta^\mu) |_{s=s_i} \end{aligned} \quad (19)$$

where $\nabla_{\theta^\mu} J_\beta(\mu_{\theta^\mu})$ is the gradient of the $J_\beta(\mu)$.

Step 8: Update the target network parameters with the Eq. (20) and (21).

$$\theta^{Q'} = \tau\theta^Q + (1 - \tau)\theta^{Q'} \quad (20)$$

$$\theta^{\mu'} = \tau\theta^{\mu} + (1 - \tau)\theta^{\mu'} \quad (21)$$

where τ is the update factor, which is taken as 0.01 in this study.

Step 9: When the number of iterations is reached, the iteration ends, otherwise skip to Step 1.

4 Case study

This section takes the Guangzhou Metro as an example, and selects the historical full load rate, platform waiting time and AFC data from May 15 to 21, 2017, which including AFC data with a total of 24.45 million travel records, for case analysis to verify the feasibility of the method.

4.1 Parameter setting

The parameters of the optimization model consist of the perceived rate of change for each passenger's travel time, number of transfers, waiting time, and route congestion degree. The JND model parameters initialized by each passenger are the same, and the values are shown in Table 1 (Ceder, & Jiang, 2020). The parameters are included in the DDPG algorithm and their values are shown in Table 2. After 50 iterations of DDPG learning, the average reward value reaches convergence, and the changes are shown in Figure 1, which is consistent with the convergence situation in reference (Hossny, Iskander, & Attia, 2021).

Table 1. Initial parameters of JND-based lexicographic preference model.

Symbol	β_{trans}^{JND}	β_{time}^{JND}	β_{wait}^{JND}	β_{load}^{JND}
initial value	0.3	0.3	0.2	0.3

Table 2. DDPG parameters.

Parameter name	Parameter value	Parameter name	Parameter value
Maximum training period	100	discount factor	0.9
Maximum training step size	10	batch size	32
Actor network initial learning rate	0.001	Maximum capacity of experience pool	10000

Parameter name	Parameter value	Parameter name	Parameter value
Critic network initial learning rate	0.002	Parameter update rate	0.01

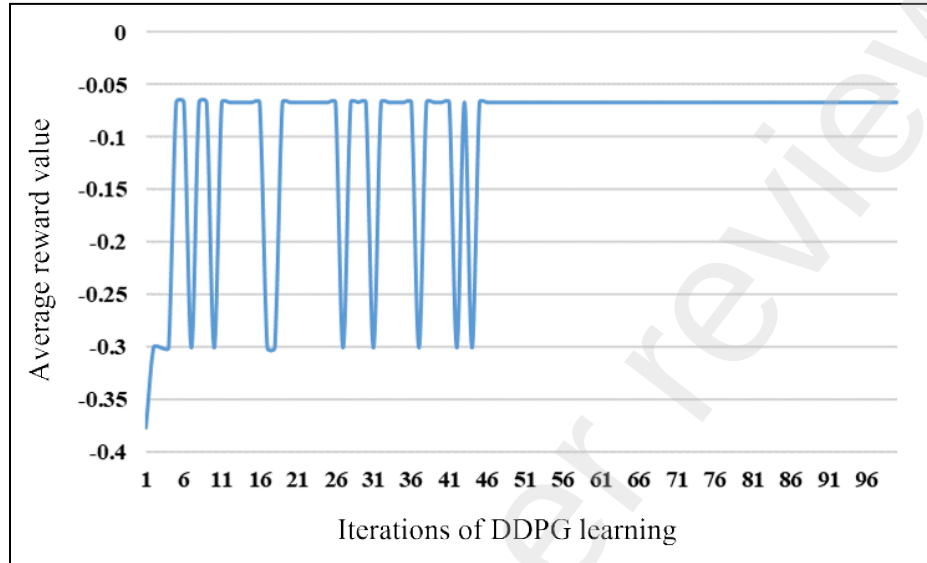


Figure 1. DDPG reward value change curve.

4.2 Identification result of passenger preferences

Due to the large number of OD pairs and frequent passengers in the entire network, without loss of generality, this section selects the multi-path non-transfer OD from Tiyu Xilu Station to Guangzhou East Railway Station during the evening peak hours and the multi-path transfer OD from Dashi Metro Station to Sun Yat-sen University Station during the morning peak hours to illustrate the passenger travel preference identification method. Since the AFC data do not contain the travel route information of passengers during their actual travel, this study adopts the individual passenger route selection model to calculate the passenger travel route (Wang, & Sun, 2018).

(1) OD data from Tiyu Xilu Station to Guangzhou East Railway Station during evening peak hours

A total of 580 passengers in this OD travel not less than 3 times on weekdays. Based on the analysis of passenger travel patterns during different time periods, as shown in Table 3, 79% of the passengers have three or more trips in one time period, and 17% of the passengers gather in the evening peak hours, and the distribution of travel time periods is relatively concentrated. In Figure 2, when the parameter gamma of kernel function is 0.5 and the number of cluster k is 3, the contour coefficient and CH fraction are the

highest, and the clustering effect is the best. So, passengers are clustered into 3 categories. Table 4 shows the index values of the cluster centers, where $p_{q_i}^{c_k}$ represents the preference of this type of passengers for label q_i . Figure 3 shows the boxplot distribution of different indicators.

Table 3. Passenger distribution at different periods of TiYu Xilu -Guangzhou East Railway Station.

Time period	The number of passengers	Passenger distribution proportion
early departure (5:00-7:00)	14	2%
morning peak (7:00-9:00)	153	26%
low peak (9:00-17:00)	125	22%
evening peak (17:00-19:00)	96	17%
late departure (19:00-24:00)	69	12%
decentralized time period	123	21%

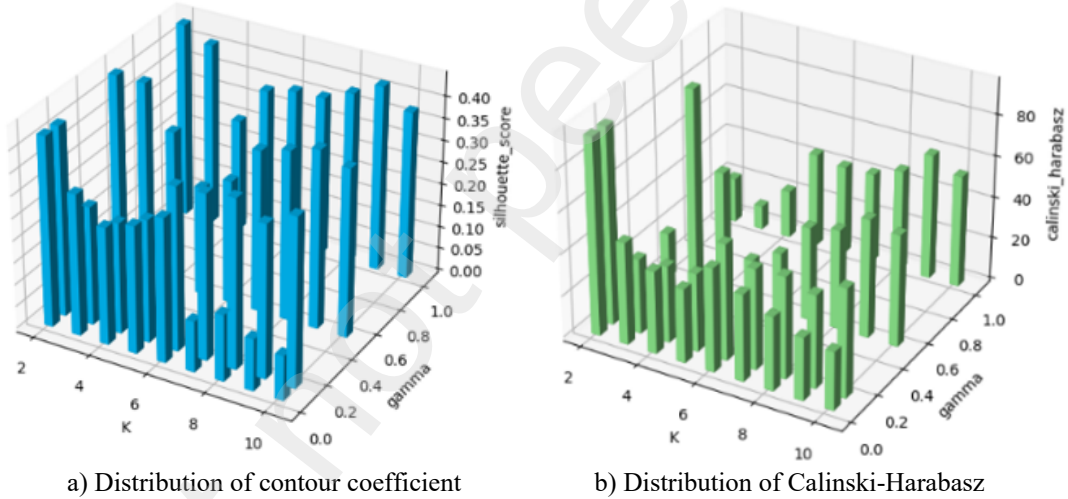


Figure 2. Spectral clustering effect diagram.

Table 4. Clustering results of passengers from TiYu Xilu Station to Guangzhou East Railway Station in the evening peak hours.

Classification c_k	1	2	3
least travel time preference $p_{travel\ time}^{c_k}$	88.94	88.69	125.22
least transfers preference $p_{transfer\ time}^{c_k}$	100	100	100
least crowded preference $p_{congestion\ degree}^{c_k}$	149.41	108.64	97.97
least wait time preference $p_{wait\ time}^{c_k}$	101.06	121.3	109.78
Proportion	31%	41%	28%

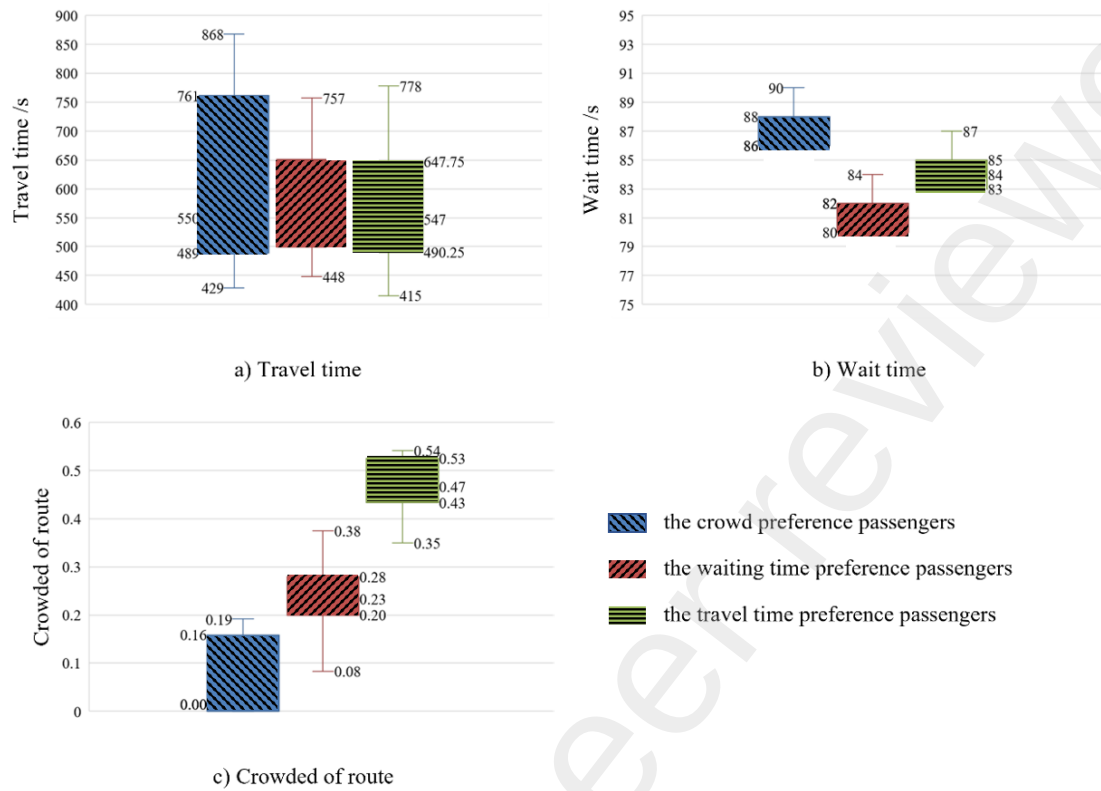


Figure 3. Boxplot of various types of passengers.

The first type of passengers are the crowd-preference passengers. This route is set in an OD without transfer, where passengers have no preference on whether to transfer or not. The preference degree of "least crowded" is 149.41 which is greater than 100. It indicates that this type of passengers has strong preference for this feature. The preference degree of "least travel time" is 88.94 which is less than 100, indicating that this type of passengers has weak preference for this feature. Tags are sorted by preference, and passenger preference is $cong > wait > time$. Further analysis on the passenger preference degree of different factors in the class shows that 84% of the passengers have the preference degree of the influencing factors as $cong > wait > time$, while for the remaining 16%, the preference degree of the influencing factors is $cong > time > wait$, which is in line with the overall preference degree.

The second type of passengers are the waiting time-preference passengers. The preference of this type of passenger is $wait > cong > time$. Further analysis on the passenger preference degree of different

factors in the class shows that the preference degree of all passengers' influencing factors is $wait > cong > time$, which is in line with the overall degree of preference.

The third type of passengers are the travel time-preference passengers. The preference of this type of passenger is $time > wait > cong$. Further analysis on the passenger preference degree of different factors in the class shows that 84% of the passengers have the preference degree of the influencing factors as $time > wait > cong$, while the remaining 16% passengers, the preference degree of their influencing factors is $time > cong > wait$, which is also in line with the overall preference degree.

(2) OD data from Dashi Metro Station to Sun Yat-sen University Station during morning peak hours

In this OD, there are 1078 passengers who travel not less than 3 times on weekdays. According to the analysis of these passengers during different time periods, 95.36% of the passengers have three or more trips in one time period, and 57.61% of the passengers gather in the morning peak hours, the distribution of travel time is relatively concentrated. Passengers from Dashi Metro Station to Sun Yat-sen University Station during the morning peak hours are clustered into 2 categories. Table 5 shows the index values of the cluster centers. The travel preferences of passengers in each type are as shown in Table 6.

Table 5. Clustering results of passengers from Dashi Metro Station to Sun Yat-sen University Station in the morning peak hours.

Classification c_k	1	2
least travel time preference $p_{travel\ time}^{c_k}$	50.33	158.78
least transfers preference $p_{transfer\ time}^{c_k}$	50.33	158.78
least crowded preference $p_{congestion\ degree}^{c_k}$	146.47	47.99
least wait time preference $p_{wait\ time}^{c_k}$	50.33	158.78
Proportion	29%	71%

Table 6. Analysis Results of Passenger Preferences from Dashi Metro Station to Sun Yat-sen University Station in the morning peak hours.

c_k	Passenger type	Passenger preference
1	Route Congestion Preferring Passengers	congestion degree $>$ travel time

c_k	Passenger type	Passenger preference
2	Travel Time Preference Passengers	travel time > congestion degree

Through analyzing the two ODs and two time periods mentioned above, the result shows that the passenger travel preference identification method is suitable for various types of ODs. This method can effectively identify passenger travel preferences in different ODs and time periods, and the identified result accurately reflects the actual travel characteristics of passengers. Furthermore, the classification of passengers in other ODs and time periods is similar.

4.3 Comparison analysis

4.3.1 Comparative analysis of individual passenger route recommendation

Taking a passenger from Tiyu Xilu Station to Guangzhou East Railway Station as an example, the proposed method identifies the passenger's travel preference. The traditional logit model, JND model, and JND model based on reinforcement learning optimization are used for route recommendation, respectively. The values of each parameter in the traditional logit model originate from the reference (Zhao, 2019). The initialization parameters of the JND model are shown in Table 1. If the recommended route is consistent with the passenger's actual travel route, the recommendation is correct. The recommendation accuracy of each model is shown in Table 7.

Table 7. Recommendation accuracy of different models.

Model	Total number of trips	Recommended correct times	Accuracy
traditional logit model	30	19	63.33%
JND Model		22	73.33%
JND Model based on Reinforcement Learning		28	93.33%

In Table 7, the accuracy of the JND model is higher than that of the traditional logit model, indicating better recommendation. The JND model based on reinforcement learning can reflect the difference in passenger sensitivity to different attribute changes, and thus improves the recommendation accuracy of the model. The optimized model parameters are shown in Table 8. Passengers have different perceivable

change ratios for different attributes, indicating that passenger sensitivity to changes in different attributes is various. Among them, the passenger has the largest perceivable change ratio to the route congestion attribute, indicating that he is not particularly sensitive to the change of the route congestion attribute. While the passenger's perceivable changes ratios for the time attribute and the transfer attribute are small, indicating that he is more sensitive to changes in the number of transfer and time.

Table 8. Parameters in the JND-based lexicographic preference model.

Symbol	β_{trans}^{JND}	β_{time}^{JND}	β_{wait}^{JND}	β_{load}^{JND}
value	0.259	0.481	0.282	1

4.3.2 Comparative analysis of route recommendation results for group passengers

Some passengers are randomly selected to verify the effectiveness of the model, and the recommendation accuracy of each passenger in each model, as well as the JND parameters optimized by reinforcement learning, is obtained. The results are presented in Table 9 and Figure 4. These passengers have different travel OD distributions and travel time distributions, and the distribution of travel times is also relatively broad, which is representative to a certain extent.

Table 9. Comparison of the accuracy in each algorithm.

Passenger serial number	Number of trips	Traditional logit model	JND model	JND Model based on Reinforcement Learning				
					β_{wait}^{JND}	β_{trans}^{JND}	β_{time}^{JND}	β_{load}^{JND}
1	30	50%	73.3%	93.33%	0.756	0.822	0.604	0.644
2	15	46.67%	100%	100%	0.436	0.550	0.702	0.295
3	7	57.14%	71.43%	100%	0.247	1.000	0.114	0.404
4	13	84.62%	92.31%	92.31%	0.845	0.167	0.189	0.628
5	10	30%	90%	100%	0.673	0.337	0.664	0.455
6	9	33.33%	66.7%	88.89%	0.548	0.105	0.256	0.516
7	12	41.67%	75%	83.3%	0.847	0.862	0.396	0.107
8	11	81.82%	90.91%	100%	0.232	0.995	0.031	0.349
9	24	100%	87.5%	100%	0.365	1.000	0.015	0.610
10	10	60%	100%	100%	0.190	0.200	0.123	0.525
average value		59%	85%	96%				
variance		0.054	0.015	0.004				



Figure 4. Algorithm accuracy fluctuation diagram.

Table 9 clearly demonstrates that the model proposed in this study performs better than the other models, with a consistent 80% accuracy rate, which is 37% higher than that of the traditional logit model and 11% higher than that of the model without reinforcement learning optimization. The perceived change rates of different attributes for different passengers exhibit significant variations after reinforcement learning optimization, indicating that different passengers possess different levels of sensitivity towards changes in the same attribute, and passenger may also have varying sensitivity towards changes in different attributes (Xu, & Xie, 2022). At the same time, the proposed model is relatively stable in Figure 4. For different passengers, the traditional logit model has large fluctuations in recommendation accuracy and a large variance. However, the model in this study has relatively stable recommendation accuracy fluctuations, and its accuracy variance is much smaller than the traditional logit model.

Through the above case analysis, it can be found that the model proposed in this study better meets the individual needs of different passengers, and recommend more accurate routes that meet passenger travel preferences.

5 Conclusions

This study develops a route recommendation method for frequent passengers in subway based on passenger preference ranking. In terms of travel preference analysis of frequent passengers, by defining lexicographical passenger preference and constructing passenger preference labels, spectral clustering algorithm is used to classify the passengers traveling within the same OD and time period, and then extract and describe different types of passenger preferences. Secondly, to consider the passenger preference order of different attributes into the route selection model, a lexicographical preference model based on JND is preliminary match between route recommendation with passenger preference. Further, considering passenger sensitivity in different attributes, reinforcement learning method optimizes the model parameters to improve the induction effect. A case of Guangzhou metro network is used to verify the effectiveness of the above methods from the individual and group levels. The results show that compared with the traditional logit model and JND model without parameter optimization, the accuracy of the proposed model is 30% and 20% higher for individual passenger recommendation, and it is 96%, 37% and 11% higher for group passenger recommendation. Meanwhile, the variance of recommendation of this model is much smaller than the traditional logit model, meaning that the model in this study performs better in meeting personalized passenger travel needs.

There exist several limitations in the proposed method to be studied in the future work. First, the travel route information involved in the acquisition of passenger preference is based on the presumption of AFC data. In the presumption model, the lack of travel time difference for passengers at different entrances and exits of the station may cause inconsistency with the actual passenger travel routes, and thus leads to inaccurate recommendation and guidance information for passengers. Second, the model in this study only considers passenger perception difference in route attribute preference, while does not consider the impact of socioeconomic passenger attributes on route selection. Future research direction of this study is to introduce the influence of passenger age, gender, income and other attributes to the passenger route recommendation. Third, the sample size of the model in the case analysis is small, so the model accuracy for large sample size needs to be verified. These will be the focuses of the future studies.

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